Yiming Yang

Ph.D. Candidate in ECE department, Michigan Tech

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EDUCATION

Ph.D. in Electrical and Computer Engineering Expected Jun 2026

Michigan Technological University, Houghton, MI, USA

- Dissertation: Robust Autonomous Driving in Winter Weather: Perception, Planning, and Control Under Adverse Conditions

Master in Control Science and Engineering Aug 2017 - Jul 2020

Chang'an University, Xi'an, China

- Thesis: Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons

Bacholar in Automation Aug 2013 - Jul 2017

Chang'an University, Xi'an, China

- Graduated from the "Excellent Engineer" honors program

RESEARCH INTERESTS

- **Stochastic and Data-Driven Optimal Control** (MPPI, Distributed Control, RL)
- **Cooperative Automation in Mixed Traffic and Multi-Agent Systems**
- **Resilient Perception and Planning in Adverse Environmental Conditions**
- **Physics-Informed Machine Learning for Vehicle Dynamics**

TEACHING EXPERIENCE

Graduate Teaching Assistant *Course: Robotic Systems Enterprise (Controls and Mathworks Team)* Jan 2023 - present

- Lead weekly group meetings and lab sessions for 6-10 undergraduate students.
- Manage team sprints by defining SMART goals, tracking the development process, and providing continuous technical feedback. Hosted office hours to assist students.
- Guide the team in developing and deploying the steering, brake and propulsion controllers using the Model-Based Design (MBD) methodology.
- Guide the team in designing integrated control logic and safe autonomy state machines with Stateflow.
- Supervise code generation (dSPACE ConfigurationDesk) and HIL/VIL testing (dSPACE ControlDesk, MicroAutoBox III).

Graduate Teaching Assistant *Course: Robotic Systems Enterprise (Safety and Testing Team)* Aug 2024 - present

- Lead weekly group meetings and lab sessions for 8-12 undergraduate students.
- Design vehicle functional safety standards and testing procedures; guide teams in writing key safety documents (e.g., SOTIF, Safety Validation Report).
- Guide the team in developing and deploying decoupled lateral and longitudinal controllers for safe vehicle functions validation.
- Lead the team in conducting safety and testing procedures, performing data collection and analysis, and writing technical reports.

RESEARCH EXPERIENCE

Ph.D. Candidate, Michigan Technological University Aug 2021 - present

Proposal: Robust 3D Human Pose Estimation and Tracking in Adverse Winter Conditions through Thermal and VIS Fusion Under Review

- Role: Lead Author and Researcher.
- Sponsor: Sony Research Award Program.
- Description: Addresses a critical gap in 3D-HPE by proposing the creation of the first multi-modal dataset for heavy snow and a novel thermal-VIS fusion approach to overcome SOTA limitations. Submitted Sep 2025.

SAE Autodrive Challenge II Jan 2023 - present

- Role: Graduate Teaching Assistant.
- Sponsor: General Motors & SAE International.
- Contributions: Architected the vehicle control interface with dSPACE MicroAutoBox III; designed and implemented the ROS-dSPACE communication pipeline; developed a safety-critical autonomy state machine with robust failure handling and safe-exit mechanisms; led HIL and VIL testing and subsystem validation on dyno and in real world; Authored DVP&R documentation to define validation scope and track system reliability.

The Railroad Crossing Vehicle Warning (RCVW) system Aug 2021 - Dec 2022

- Role: Graduate Research Assistant.
- Sponsor: Federal Railroad Administration (FRA).
- Contribution: Led the V2X system implementation, including the configuration of VBS/RBS/RSU hardware and software, railroad crossing map creation, and field testing; developed a HIL virtual demonstration system to replicate real-world scenarios; created and presented large-scale virtual demonstrations of the RCVW application for Connected Vehicles using PTV VISSIM for the Rail Learning System (RLS) and FRA website.

Integration of Unmanned Aerial Systems Data Collection Into Day-to-Day Usage for Transportation Infrastructure – A Phase III Project Nov 2021 - Dec 2022

- Role: Graduate Research Assistant.
- Sponsor: Michigan Department of Transportation (MDOT).
- Contributions: Executed detailed data labeling on the Williamston Road (I-96) traffic corridor dataset to train a machine-learning model for autonomous vehicle counting; subsequently analyzed the traffic data to generate OD insights and validate vehicle counts.

Microscopic Traffic Safety Analysis and macro-level transportation analysis Aug 2021 - Dec 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Conducted a microscopic safety analysis on the Waymo dataset by visualizing agent dynamics against road structures, extracting lane geometries to identify intersection conflict points, and generating lane polygons for precise map-matching; additionally, performed a macro-level transportation analysis using the Weijo dataset to conduct an OD study, identifying key mobility trends and peak-hour congestion zones.

From OSM to VISUM: Automated Infrastructure Generation for Transportation Planning. Aug 2021 - Dec 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Developed an automated pipeline to convert OpenStreetMap data (nodes, ways, tags) into simulation-ready PTV VISUM networks while ensuring network connectivity, generated Points of Interest and automated zone divisions, and exported the final topology into VISUM-compatible formats

(.net/.ver). This automation reduced manual network-building time from days to minutes, enabling rapid scenario testing.

From OSM to VISSIM: Automating Large-Scale Traffic Network Modeling Aug 2021 - Dec 2022

- Role: Graduate Research Assistant.
- Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
- Contributions: Developed an automated pipeline converting OSM data into high-fidelity PTV VISSIM models; interpreted road class, lane counts, and speed limits to model MUTCD-compliant intersections and utilized spline-based connectors for precise vehicle paths. The process generated native (.inpx) files, enabling rapid creation of large-scale, high-resolution traffic models.

Master's Student, Chang'an University Aug 2017 - Jul 2020

Research on Construction and Site Testing Technology of Closed Test Environment for Auto-driving Electric Vehicles Jul 2018 - Jun 2020

- Role: Graduate Research Assistant.
- Sponsor: National Key Research and Development Program.
- Contributions: Led hardware and software adaptation to transition the autonomous system from gasoline to EV architecture; developed an single-chip microcomputer based vehicle controller, and engineered and calibrated precise drive-by-wire algorithms for steering, propulsion, and braking.

QLTD Intelligent and Connected Expressway Testbase construction April 2019 - Sep 2020

- Role: Graduate Research Assistant.
- Sponsor: QiLu Transportation Development Group and Chang'an University.
- Contributions: Implement V2X systems for both control - an adaptive IDM and a cellular-based remote control system; and safety - developed a V2X-based forward collision warning system. All functionalities were validated through real-world highway platoon tests using V2V (DSRC/C-V2X).

Key Technologies for Driverless Commercial Vehicle Control Systems Aug 2018 - Jun 2020

- Role: Graduate Research Assistant.
- Sponsor: Key Projects of Key Research and Development Programs in Shaanxi Province.
- Contributions: Designed and implemented a by-wire motion-control system (throttle, brake, steering) using incremental PID algorithms; deployed trajectory tracking algorithms including Pure Pursuit and MPC; additionally, engineered a safety-critical remote controller with override/E-stop functionality and power supply system for all onboard electronics.

PUBLICATIONS

1. Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection In Prep.

Yiming Yang, Jeremy P. Bos

2. Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion Under Review

Yiming Yang, Jeremy P. Bos - Submitted to IEEE Transactions on Intelligent Vehicles (T-IV), May 2025

3. Aggressive Autonomous Control on Snow and Ice Apr 2025

Yiming Yang, Jeremy P. Bos

SAE International Journal of Advances and Current Practices in Mobility (10.4271/2025-01-8040)

4. Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons Nov 2019

Haigen Min, Yiming Yang, Yukun Fang, Pengpeng Sun, Xiangmo Zhao, 10.1109/ACCESS.2019.2952049

TALKS & PRESENTATIONS

1. Aggressive Autonomous Control on Snow and Ice Apr 2025
Conference proceedings talk, World Congress Experience (WCX), 2025, Detroit, MI, USA

2. Deep Deterministic Policy Gradient based Cooperative Platoon Logitudinal Control Strategy Jan 2021
Conference presentation, Transportation Research Board (TRB), 2021, Virtual

3. Research on Automatic Emergency Line Change Control Strategy Based on MPC Jun 2019
Conference presentation, World Transport Convention (WTC), 2019, Beijing, China

PATENTS

1. **Autonomous vehicle controllers** (NO: ZL 2019 2 0786357.5)
2. **Autonomous vehicle platoon controller** (NO: ZL 2019 2 0784716.3)
3. **Unmanned vehicle power system and unmanned vehicle** (NO: ZL 2019 2 1037928)

SELECTED HONORS & AWARDS

1. 2nd Place, Concept design report & SRS - SAE/GM AutoDrive Challenge II (Year 4) Jun 2025
2. 2nd Place, Highway Challenge - SAE/GM AutoDrive Challenge II (Year 2) Jun 2023
3. 3rd Place, Overall Dynamics - SAE/GM AutoDrive Challenge II (Year 2) Jun 2023
4. Excellent graduate of Chang'an University Jun 2020
5. Excellent graduate of Chang'an University Jun 2020
6. First-class Academic Scholarship 2017-2018 Nov 2018
7. 2017-2018 University-level Outstanding Graduate Student Nov 2018
8. Best Technology Award of 2018 i-VISTA Auto Driving Challenge Aug 2018
9. First Prize of 2018 i-VISTA Auto Driving Challenge - Urban Traffic Scene Challenge Aug 2018
10. Second Prize of 2018 China Robot Competition - General Service Robot Project Aug 2018
11. First-class academic scholarship 2016-2017 Nov 2017
12. University-level first prize: Electric vehicle charging pile and operation management platform, Team award Jan 2017

SKILLS

Computational: Python, C/C++, MATLAB/Simulink for algorithm development and data analysis
Control Systems: Real-time control design and implementation (dSPACE, ConfigurationDesk, ControlDesk)
Traffic Simulation: PTV Vissim, PTV Visum
Robotics: ROS, GAZEBO, dSPACE integration with ROS
Hardware: CAN bus analysis, CAD (SOLIDWORKS), and PCB design (Altium Designer)